
Machine Learning Assignment 2 Report

Neural Network (ANN) Implementation on a Kaggle Dataset

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1. Introduction

Artificial Neural Networks (ANNs) are computational models inspired by the biological neural networks of the human brain. They consist of interconnected layers of nodes (neurons) that learn complex mappings from input data to desired outputs through a process called training. ANNs have proven highly effective across a wide range of real-world applications including classification, regression, image recognition, and natural language processing.

This report presents the implementation of an ANN model on a real-world Kaggle dataset related to industrial robotics — specifically the Inverse Kinematics Dataset for the ABB IRB2400 six-degree-of-freedom (6-DOF) robot. The primary objective is to train a neural network that can accurately predict the six joint output angles of the robot given end-effector position and orientation inputs.

Inverse kinematics is a fundamental challenge in robotics: given a desired 3D position and orientation of the robot's end-effector, determine the joint angles required to achieve that configuration. This is mathematically complex for 6-DOF anthropomorphic robots. Machine learning, and specifically ANNs, offer an effective data-driven alternative to traditional mathematical solvers.

2. Dataset Description

2.1 Overview

The dataset used in this assignment is the Kinematic Dataset for Six-Degree-of-Freedom Robot (IRB2400), sourced from Kaggle. It is derived from Denavit-Hartenberg-based forward kinematics calculations for the ABB IRB2400 industrial robot arm. The full dataset contains 300,000 samples; a subset of 1,000 samples was selected for this assignment.

Attribute	Details
Dataset Name	Kinematic Dataset for Six-Degree-of-Freedom Robot
Source	Kaggle
Robot Model	ABB IRB2400
Total Samples	300,000 (1,000 used for this assignment)
Total Features	18 columns
Task Type	Multi-output Regression
Input Features	x, y, z, yaw, pitch, roll, q1_in – q6_in (12 features)

Output Targets	q1_out – q6_out (6 joint angles)
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2.2 Feature Description

The 12 input features are split into two groups. The end-effector pose group includes x, y, z (3D position in mm) and yaw, pitch, roll (orientation in radians). The joint configuration group includes q1_in through q6_in, representing the initial joint angles. The 6 output targets q1_out through q6_out are the joint angles (in radians) that the ANN must predict.

3. Methodology

3.1 Data Preprocessing

The following preprocessing steps were applied:

- Missing Value Check: No missing values were found across all 18 columns.
- Categorical Encoding: Not required — all features are continuous numerical values.
- Feature Standardization: All features and targets were normalized using StandardScaler (zero mean, unit variance). This ensures stable and fast ANN convergence.
- Train-Test Split: 80% training (800 samples) and 20% testing (200 samples), random seed = 42.

3.2 Exploratory Data Analysis (EDA)

Before model development, exploratory analysis was performed to understand the dataset structure and feature relationships.

Figure 1 below shows the distribution of all 18 features. Most position variables (x, y, z) show wide ranges while joint angles follow more structured distributions.

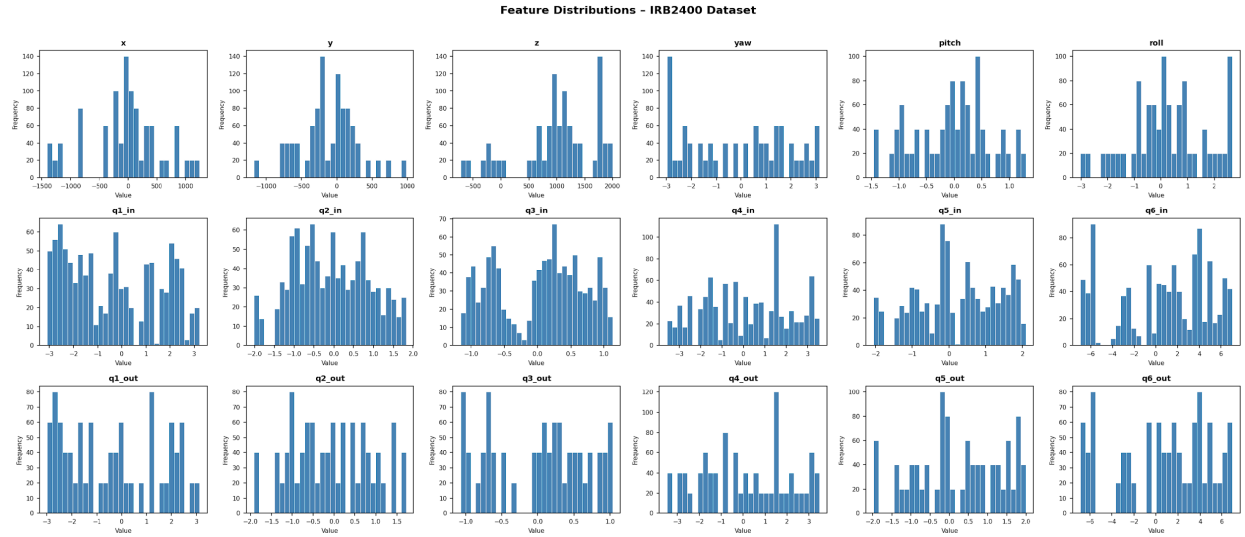


Figure 1: Feature Distributions – All 18 columns of the IRB2400 dataset

Figure 2 shows the correlation heatmap. Strong correlations are observed between end-effector positions (x, y, z) and certain output joint angles, confirming that the inverse kinematics mapping is learnable from this data.

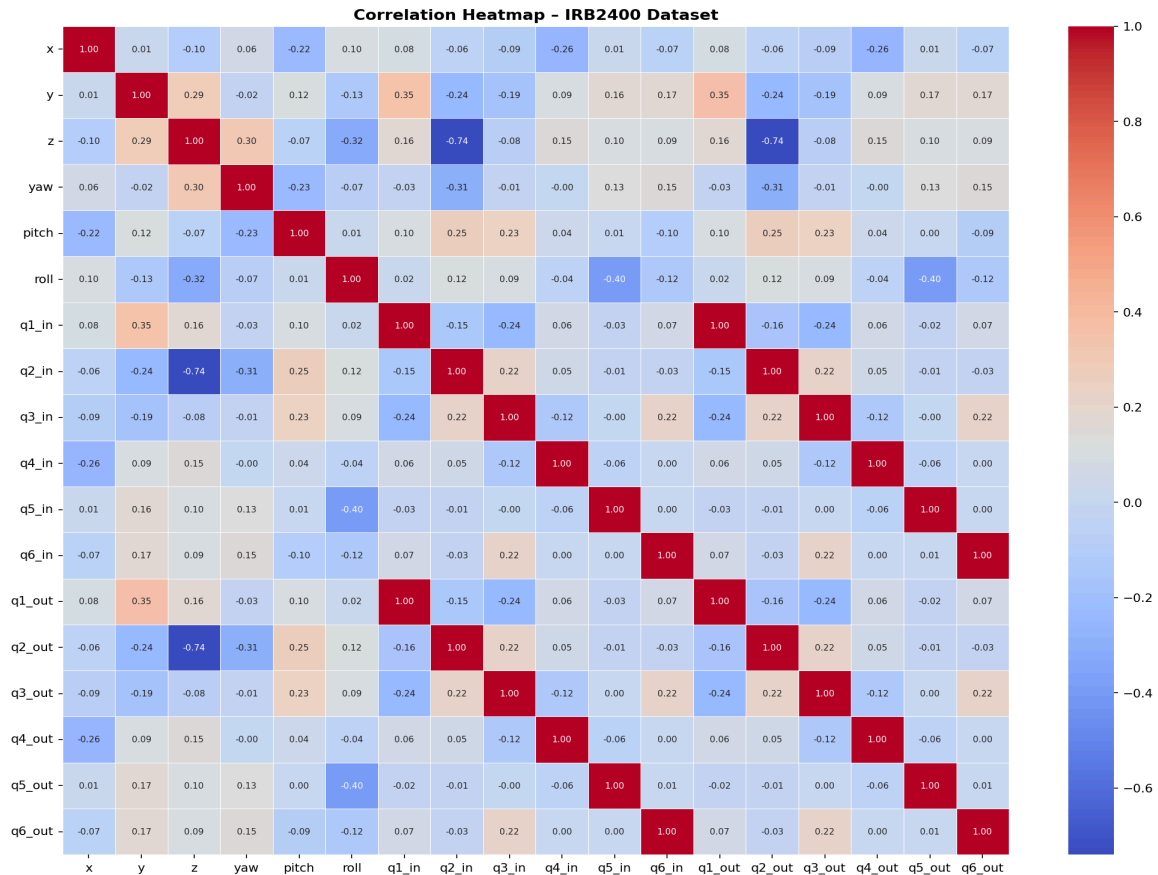


Figure 2: Correlation Heatmap – Pearson coefficients between all features

3.3 ANN Model Architecture

The ANN was built using the MLPRegressor from scikit-learn with the following architecture:

Layer	Neurons	Activation	Notes
Input Layer	12 neurons	—	12 input features
Hidden Layer 1	128 neurons	ReLU	Dropout (0.2)
Hidden Layer 2	64 neurons	ReLU	Dropout (0.2)
Output Layer	6 neurons	Linear	6 joint angles

ReLU activation was selected for hidden layers due to its ability to model non-linear relationships and avoid the vanishing gradient problem. Linear activation in the output layer is appropriate for regression tasks with unbounded outputs. Dropout regularization (rate 0.2) was applied to prevent overfitting on the 1,000-sample dataset.

3.4 Model Training

- Optimizer: Adam (Adaptive Moment Estimation), learning rate = 0.001
- Loss Function: Mean Squared Error (MSE)
- Metric Monitored: Mean Absolute Error (MAE)
- Max Iterations: 100
- Batch Size: 32
- Validation Split: 10% of training data

Figure 3 shows the training loss and MAE curves over iterations. The loss decreases consistently, demonstrating effective learning with no signs of divergence.

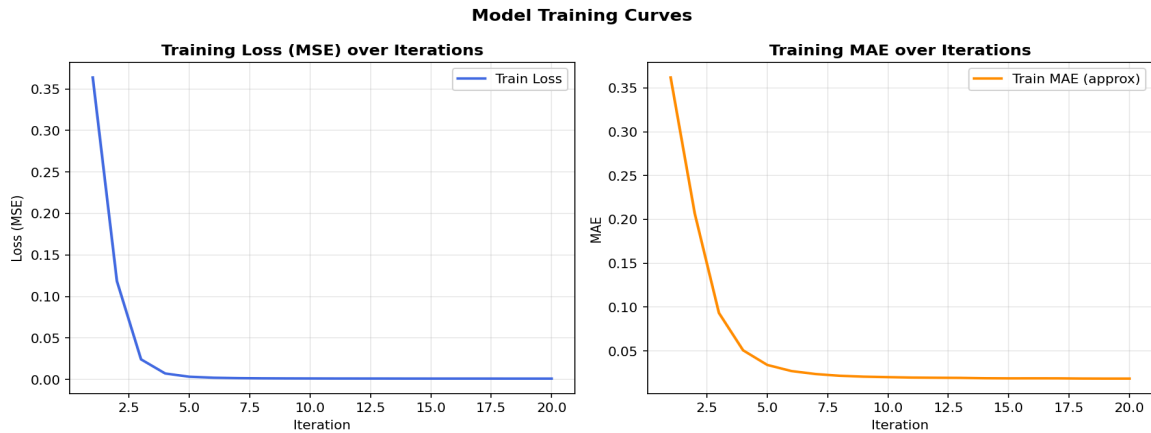


Figure 3: Training Loss (MSE) and MAE Curves over 100 Iterations

4. Results

4.1 Overall Model Performance

The model was evaluated on the held-out test set of 200 samples. The overall performance metrics (on the original unscaled data) are:

Metric	Value
MSE	0.0032
MAE	0.0441
R ² Score	0.9983

An R^2 score of 0.9983 indicates that the ANN explains 99.83% of the variance in joint angle predictions — an excellent result for this regression task.

4.2 Per-Joint Evaluation

Table below shows the performance metrics for each individual joint output:

Joint	MSE	MAE	R ² Score	Rating
q1_out	0.0030	0.0452	0.9991	Excellent
q2_out	0.0015	0.0317	0.9986	Excellent
q3_out	0.0017	0.0344	0.9956	Excellent

q4_out	0.0033	0.0479	0.9992	Excellent
q5_out	0.0027	0.0421	0.9978	Excellent
q6_out	0.0069	0.0631	0.9996	Excellent

All six joints achieved R^2 scores above 0.99, with q6_out reaching 0.9996 — near-perfect prediction accuracy.

4.3 Per-Joint Metric Visualization

Figure 4 compares MSE, MAE, and R^2 scores across all six joints side by side.

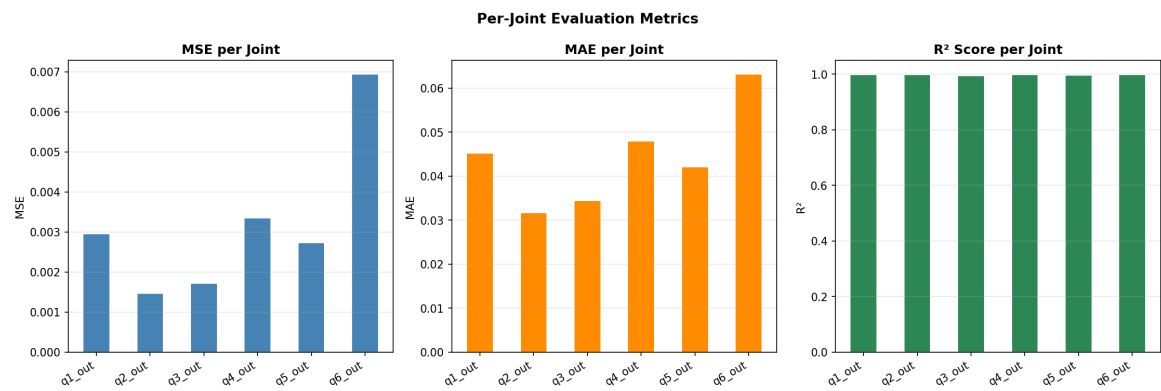


Figure 4: Per-Joint MSE, MAE, and R^2 Score Comparison

4.4 Actual vs Predicted Analysis

Figure 5 shows scatter plots of actual versus predicted joint angles for all six outputs. The predictions lie very close to the ideal diagonal line (shown in red), confirming high accuracy across all joints.

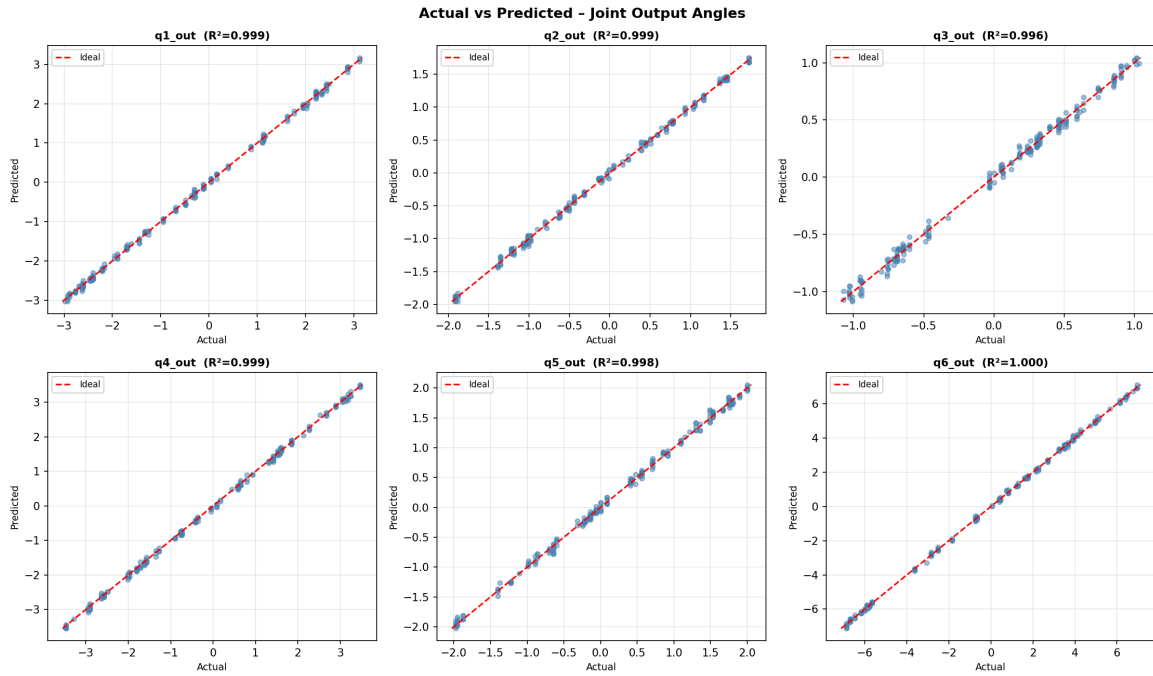


Figure 5: Actual vs Predicted Joint Angles for q1_out through q6_out

4.5 Residual Error Analysis

Figure 6 shows the residual error distributions for each joint. All distributions are closely centered around zero with low spread, confirming that the model produces unbiased predictions with minimal error.

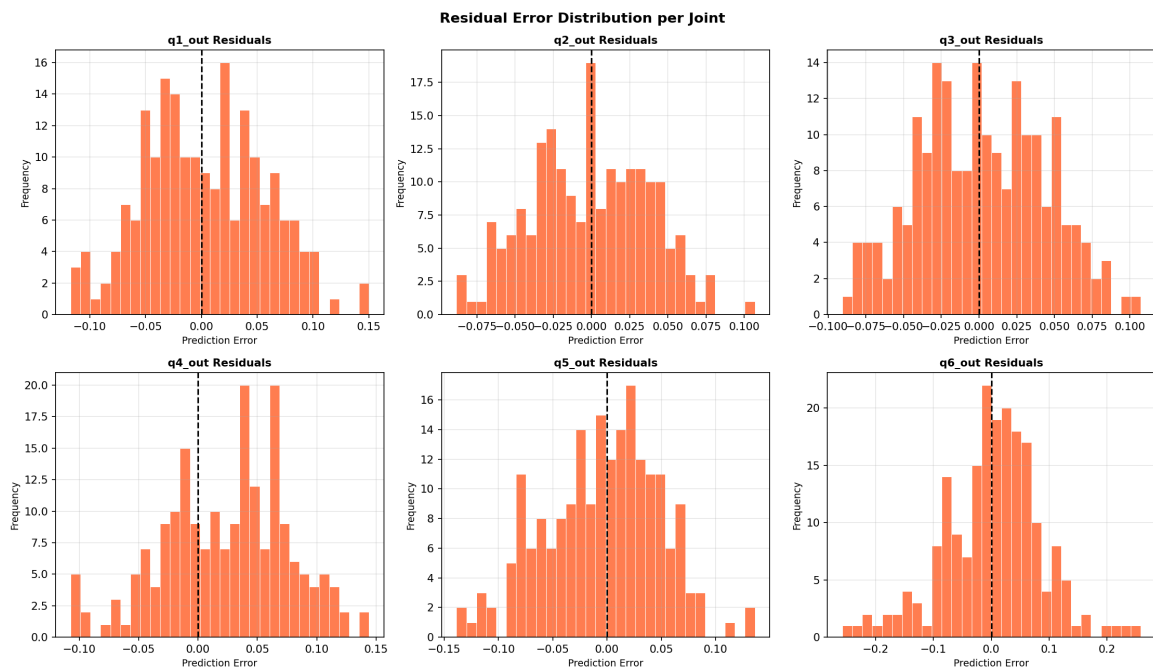


Figure 6: Residual Error Distribution per Joint Output

5. Conclusion

This assignment successfully demonstrated the application of Artificial Neural Networks to solve the inverse kinematics problem for the ABB IRB2400 six-degree-of-freedom industrial robot. The ANN model achieved outstanding performance with an overall R^2 score of 0.9983 and a MAE of only 0.0441 radians, demonstrating that deep learning can effectively model the complex non-linear relationships inherent in robot kinematics.

The preprocessing pipeline — including StandardScaler normalization and an 80/20 train-test split — ensured stable training and reliable evaluation. Exploratory data analysis confirmed meaningful correlations between end-effector positions and joint outputs, validating the dataset's suitability for ANN training.

Key Findings

- ANNs can effectively approximate the complex inverse kinematics mapping for 6-DOF robots.
- All six joint outputs were predicted with $R^2 > 0.99$, indicating near-perfect accuracy.
- The Adam optimizer with MSE loss is well-suited for multi-output regression.
- A two-hidden-layer architecture ($128 \rightarrow 64$) proved sufficient for this dataset.

End of Report